Harshana Lakshara Fernando

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Summary

aBackend-focused Computer Science Engineering undergraduate with a rigorous foundation in **System Design**, **Distributed Systems**, and **Cloud-Native Engineering**. Proven experience architecting high-performance, fault-tolerant applications using **Microservices**, **Serverless Architectures**, and **Event-Driven Design** with **Java (Spring Boot)**, **Go**, **AWS Lambda**, and **Kafka**. Validated through the delivery of high-throughput systems handling massive load, seeking to solve complex **concurrency** and **low-latency** challenges within high-velocity engineering teams.

Education

B.Sc. (Hons) in Engineering – Computer Science and Engineering *2022 – 2026*
University of Moratuwa, Sri Lanka

G.C.E. Advanced Level (Physical Science Stream) *2019*
St. John's College, Panadura, Sri Lanka
Results: 3 A's (Combined Mathematics, Physics, Chemistry) **District Rank:** 17

Experience

Software Engineer — Trading Systems *Sep 2026 – Present*
CAL | Colombo, Sri Lanka

- Building CAL's in-house electronic trading platform: a direct gateway to the **Colombo Stock Exchange (CSE)** plus the dealing, operations and testing tools around it.
- Led CSE's **188**-case **exchange certification**, building a Certification Runner that drives each case and explains every field difference against the expected result, backed by durable message logs and a CI-enforced test register.
- **Event-sourced CQRS** order path: a **Chronicle Queue** write-ahead journal with **PostgreSQL** as an async read model, so no database write sits between a trader and the exchange; the day's orders replay after a restart.
- **FIX 5.0 SP2** order routing and drop copy (**QuickFIX/J**), a custom **Java NIO** decoder for the **MITCH** UDP multicast feed (A/B feeds, snapshot recovery) and client-side rate limiting.
- Designed an **exchange simulator** (Python control plane, Java price-time matching engine) that replays **1,700+** real exchange conversations with **zero parity failures**, covering price bands, auctions and the trading-day schedule.
- Resilience and operations: FIX and market-data **failover** across primary and mirror hosts, sequence-number and feed-gap recovery, drop-copy **reconciliation** with an audited adopt/close path, **health-gated deployment** with rollback, and order-path changes behind default-off feature flags.
- Built the ops console (session control, reconciliation, chaos tests) and the **React** dealing screen, with pre-trade warnings for price bands, self-match and uncancellable AON orders.
- **~1,050 automated tests** (637 Java, 195 simulator, 216 console) in CI; every exchange behaviour traceable to the CSE spec or captured wire evidence.

R&D Software Engineer — KōreroCare Platform *Mar 2026 – Aug 2026*
Empathic Computing Laboratory, University of Auckland | Remote / Part-time

- Co-led backend design for the **KōreroCare** mobile-wearable sensing platform, building **Go/Gin**-based services for time-series ingestion and cross-device synchronization.
- Designed **PostgreSQL** schemas for participants, studies, devices, consent, reminders, and wearable sensor streams.
- Integrated **Keycloak OAuth2/OIDC** with **Traefik** gateway routing and Docker Compose deployment for secure internal service access.
- Developed Firebase notification and reminder flows for participant engagement and study prompts.
- Modeled an **HL7/FHIR**-aligned data schema mapping participants, devices, and wearable measurements.

Co-founder & Lead Engineer (Freelance) *Jul 2025 – Aug 2026*
ScaleX — SaaS Development Lab *Remote / Sri Lanka*

- **Stream-Native Tipping Platform:** Architecting a creator-focused SaaS to solve Sri Lankan payout barriers by integrating PayHere's aggregated gateway. Developed an OBS-native alert system using webhook-driven `notify_url` callbacks to trigger real-time on-stream overlays.
- **High-Performance Scheduling SaaS:** Engineering a mobile-first scheduling tool using Next.js 14 and Supabase Realtime for live heatmap synchronization. Implemented a custom best-slot recommendation algorithm to rank optimal meeting times based on participant overlap and duration constraints.
- **Scalable Infrastructure & DevOps:** Designed a $0-to-launch deployment strategy leveraging Vercel serverless functions and Supabase PostgreSQL with connection pooling. Established robust security protocols including Zod-based input validation and rate-limiting to ensure system resilience.
- **Strategic Compliance & Growth:** Navigating CBSL regulatory frameworks to operate as a software platform while delegating fund-handling to licensed gateways to minimize operational risk. Implementing a "freemium" viral loop strategy with branded footers and Stripe-integrated Pro tiers for monetization.
- **AgriNova:** Built a full-stack inventory and commission management platform (**Next.js**, **Prisma**, **Supabase**/**PostgreSQL**) replacing spreadsheet workflows: 5 branches, 329+ SKUs, 194K+ inventory units tracked and 860+ invoices processed (Rs. 64.4M+) with automated commission calculations. Cut repeat dashboard data-fetch latency from ∼384ms to under 1ms with React Query caching and pagination.

Software Engineer Intern (Fullstack Focus) *Dec 2024 – May 2026*

Information Systems Associates (Pvt) Ltd — Aviation Technology Company

- Engineered a distributed, high-throughput unique ID generator for Passenger Name Records (PNRs) using a Kafka-backed pool model, ensuring uniqueness and security across scaled instances while adhering to aviation domain standards.
- Architected and implemented a hybrid integration strategy by embedding a gRPC client and Kafka consumer into a legacy Java 8 monolith (AeroMart), enabling seamless, backward-compatible communication with the new AeroOrder microservices platform.
- Developed dynamic, database-driven feature toggles to control PNR routing logic, minimizing risk during the phased production rollout by allowing for granular control and immediate fallback.
- Actively contributed to the agile development lifecycle, participating in all ceremonies, performing code reviews, and collaborating with cross-functional teams to deliver production-ready features using Jira and Bitbucket.
- Integrated frontend components with backend services to deliver end-to-end features.

Keywords: Java 18, Spring Boot, React, gRPC, Kafka, REST API, PostgreSQL, Docker, Microservices, Agile.

Open Source Contributions **See all**

- **KEDA #8317** (CNCF) *Merged*
  New **onlyReadyPods** option for the Kubernetes workload scaler (count only Running and Ready pods); design agreed with a maintainer; docs keda-docs #1919 merged Nov 2026.
- **kgateway #14851** (CNCF) *Merged, Oct 2026*
  Added the **kgateway_build_info** Prometheus metric (version, revision, Go version, OS/arch) so operators can see which build each pod runs; shipped in **kgateway v2.5.0-beta.2**.
- **kubernetes-mcp-server #1482** *Merged, Oct 2026*
  Fixed flaky HTTP server startup in the **Go** test suite with a 5 s readiness poll and cleanup before failing.
- **kubernetes-mcp-server #1549** *Merged, Oct 2026*
  Test that every tool, prompt and resource across all 12 toolsets declares valid **Kubernetes RBAC** metadata; approved on the first review.
- **Kubeflow Spark Operator #3249** *In review (LGTM)*
  20 **Ginkgo** specs covering every executor pod-phase transition: **100%** of `sparkpod_metrics.go`, plus a log message fix.
- **Jaeger #9997** (CNCF) *In review*
  Found and reported (#9991) a bug in Jaeger's **MCP** trace-topology tool: reused span IDs looped and parent cycles dropped spans; fix with 6 tests.
- **kgateway #14945** (CNCF) *In review*
  statPrefix for local rate limits on the traffic policy, with route template tokens; unit and translator tests.
- **Apache SkyWalking BanyanDB #1462** *In review*
  Found and fixed (#14199, assigned to me) case-sensitive **Bearer** auth in the **MCP** HTTP server; 4 tests; milestone 0.12.0.
- **Apache SkyWalking #14194** *Bug reported, Nov 2026*
  Found and reported **MCP** server config bugs (`.env` ignored, malformed values silently accepted) with a reproduction; fix by another contributor in banyandb #1460.

- **Apache Camel K #6933** *In review*
  Apache Pulsar autoscaler for Camel K's **KEDA** trait: scaler mapping, unit and e2e tests, docs.
- **kubernetes-mcp-server #1568** *Root-caused, Oct 2026*
  Showed a reported "flaky" test was a real merge-order bug; the author confirmed and applied the suggested fix.

Projects — See more projects on my portfolio

1. AgriNova Inventory & Commission Platform (2025 | Freelance, ScaleX) **Live demo**

Designed and delivered a production-grade inventory and sales management system for a real-world client, supporting multi-branch stock operations, invoicing, receipts, and commission tracking.

- Engineered transaction-safe stock movement logic and replaced application-layer aggregation with **PostgreSQL** aggregations, reducing execution time from **∼384 ms to ∼0.02 ms**.
- Designed a multi-layer caching strategy with **Next.js** and **React Query**, reducing repeated dashboard latency from **∼703 ms to ∼0.05 ms**.
- Built a clean, dynamic admin-focused UI and validated system reliability through end-to-end testing with **Playwright**.

Keywords: Next.js, PostgreSQL, Prisma, React Query, Tailwind CSS, Playwright

2. Employee Management System (EMS) (2025 | Individual) **Architecture | Github**

Designed and built a comprehensive, cloud-native microservices application for employee and department management, featuring a full-stack architecture with a React/TypeScript frontend and Spring Boot backend.

- Architected a distributed system using **Spring Cloud** for service discovery, centralized configuration, and an API gateway, with **Resilience4j** for robust fault tolerance.
- Enabled dynamic configuration management using **Spring Cloud Bus** with **RabbitMQ** and established a robust observability stack (Zipkin, ELK, Prometheus/Grafana).
- Designed and managed a CI/CD pipeline using **GitHub Actions**, with **ArgoCD** for GitOps-based deployments to the Kubernetes cluster.

Keywords: Java 17, Spring Boot, Spring Cloud, React, TypeScript, Resilience4j, RabbitMQ, Docker, Kubernetes, GitHub Actions, ArgoCD, Prometheus, Grafana

3. GoTogether (2024 | Group) **Architecture | Live | Github**

Developed and deployed a full-stack travel assistant platform with a **Next.js** frontend (SSR/CSR, secure auth) and microservices in **Spring Boot** and **Go**, enabling AI itinerary generation via **Gemini API**.

- Implemented a **Spring Boot** social service with **Keycloak** integration for identity management and a dedicated **Go** service using **gRPC** for high-performance inter-service communication.
- Configured production-grade security with **NGINX, Certbot, and DuckDNS**, and orchestrated deployment on **Oracle Cloud Free Tier** using custom scripts and Docker optimizations.
- Built a dedicated **Auth Service using Keycloak**, managing OAuth2/OIDC flows, token issuance, and automated realm/client boot-time import scripts.

Keywords: Next.js, Go, Spring Boot, Keycloak, PostgreSQL, NGINX, Docker, Gemini API, Oracle Cloud, gRPC

4. Food Delivery Application (2024 | Group) **Github**

Built a scalable, event-driven food delivery platform inspired by systems like Uber Eats, using a modern microservices architecture. As part of the core team, I focused on designing and developing the Restaurant Service.

- Developed the Restaurant Module with secure sign-up/login using **Spring Security** and **JWT** (with extended claims).
- Built CRUD APIs for menu and item management; integrated **Redis caching** to improve performance and reduce database load, with appropriate cache invalidation strategies.
- Enabled order sync via **Kafka**: when an order is placed, an event is published to the 'order_created' topic, which the restaurant service consumes to update order details.

Keywords: Java, Spring Boot, gRPC, Kafka, Redis, PostgreSQL, WebSocket, Docker, React.js, Ant Design

5. Maison Vella – E-Commerce Admin Panel (2024 | Individual) **Architecture | Live | Github**

Comprehensive admin interface for the Maison Vella e-commerce ecosystem, facilitating centralized management of products, orders, and content assets.

- Engineered a responsive **Next.js 14** app with **Server Actions**, **tRPC**, and **ShadCN UI**, implementing **CQRS** and layered architecture.
- Integrated **Sanity** (Headless CMS) via GraphQL and managed data with **Prisma** and **Supabase** in a serverless environment.
- Secured access with **Clerk Auth** and optimized media delivery using **Cloudflare R2** and a global CDN.

Keywords: Next.js 14, Server Actions, tRPC, GraphQL, Prisma, Supabase, Sanity, Serverless, Cloudflare R2, CQRS

6. Virtual Me – 3D Animated AI Chatbot (2024 | Individual)

Created an interactive 3D virtual assistant with lifelike animations and AI-driven conversations.

- Integrated **Hugging Face** (Mixtral model) for natural language dialogue.
- Implemented **Eleven Labs** TTS for human-like voice output.
- Built immersive 3D interface with **React** and **Three.js**.

Keywords: React (Three.js), Node.js, Express.js, Hugging Face, Eleven Labs, Rhubarb, FFmpeg

Training & Certifications

CKA Training — KodeKloud, 2025 • AWS SAA-C03 Training — Udemy, 2025 • Apache Kafka Series — Udemy, 2025 • 100 Days of DevOps — KodeKloud, 2025 • Advanced OAuth Security — Udemy, 2024

Skills

Core	**Java** | **Spring Boot** | **Microservices** | Distributed Systems | System Design | **REST APIs**
Backend	Spring Cloud | Spring Security | Resilience4j | Node.js | Express.js | Gin
Messaging & APIs	**Kafka** | RabbitMQ | **gRPC** | **GraphQL** | WebSockets | Protobuf | JWT | OAuth 2.0
Data	**PostgreSQL** | MySQL | Oracle DB | Redis | Couchbase
Cloud & DevOps	AWS | **Docker** | Kubernetes | CI/CD (GitHub Actions, Jenkins)
Testing	JMeter | Gatling | Pytest | Unit Testing
Tools	Git | Gradle | Postman | Jira | Confluence

References available upon request.